Literal structural certificates for Sp(4, 4)

L054 recovery component

Theorem 1. *Each of the 496 actual subgroup representatives in the acquired numerical catalogue for the fixed current* Sp(4, 4) *companion table admits its displayed recursive structural description, with the source-qualified operation conventions below.*

The table is attributed to Thomas Connor and Dimitri Leemans, has ambient order 979200, occupies 28 native pages, and prints June 7, 2013. All 28 pages were directly inspected. This application concerns these exact retained bytes; it does not assert their identity with an earlier website edition, uniqueness of structural names, or completion of the entire article. The accepted numerical catalogue and its source-row bijection are retained inputs. No ambient subgroup census is repeated.

Actual group and named input. The accepted six-table numerical component explicitly checked that the actual complete 85-point state equals the named constructor `PSp(4,4)`. Its exact state, constructor interface, receipt, source bijection and acceptance are acquired here. Every proper representative is read from literal embedded permutations in that same state; no displayed subgroup name is substituted for its embedding. The usual named-constructor definition of the projective symplectic group is a retained software interface, not inferred from an order match.

For completeness, the source's Sp(4, 4) and that projective name agree. On a nondegenerate symplectic space over $\mathbf{F}_4$, each $T_v(x) = x + \langle x, v\rangle v$, $v \neq 0$, is a symplectic transvection. If A is central then $T_{Av} = AT_vA^{-1} = T_v$. Equality of the nonzero rank-one operators $T_{Av} - I = T_v - I$ implies $Av = \lambda v$ and $\lambda^2 = 1$. In characteristic two the latter forces $\lambda = 1$. Thus A fixes every nonzero vector and is the identity. The symplectic center is trivial, so passage to the projective quotient changes no group.

Names and source joins. The stored syntax tree preserves the displayed parentheses and punctuation. Absent parentheses, the acquired recursive naming convention is left association. A colon asserts a split extension (with nontrivial displayed action in these certificates), a dot a nonsplit extension, and a cross an actual direct product. Superscripts are recovered from native typography: in particular $2^4, 2^5, 2^6$ are elementary abelian groups, not cyclic groups of orders $24, 25, 26$. The wrapped row 22 remains $2 \times 4 : 2 \times 2 \times 2 : 2 : 2 : 3$, with its exact source text retained. The source checker requires the same row number, group order, literal name, syntax tree and acquired actual representative. Every original ambient and subgroup generator is pinned.

Complete finite certificates. The shared acquired registry supplies previously verified named atoms and extension models. New models have literal generators and finite closures. For every model-to-factor or model-to-row map, the verifier reconstructs the image of every element along the model's complete ordered-generator Cayley graph. It checks identity, all right-generator edges, injectivity and exact surjectivity onto the actual subgroup. The acquired generator-edge theorem then gives a group isomorphism: induction on words proves compatibility with multiplication by any element. No partial presentation, successful search label or comparison of orders alone is used as an isomorphism proof. The inherited byte-array representation of these permutations has the same literal composition law as the earlier accepted reader.

For an extension $H = N \star K$, literal closure and conjugation by all generators of H prove $N \lhd H$; the factor's registry map proves its name. In a split case the literal complement has trivial intersection with N and $|N||K| = |H|$, so $H = NK$. In a direct case all factor generators commute; commutation is preserved under products and inverses, so both factors commute elementwise. For a prime-index dot, the retained exhaustive test verifies that every element outside N has order different from the prime. A complement would contain such an element, giving a contradiction. Any further extension criterion used is the exact acquired literal criterion, not a negative GAP complement-search result.

For an elementary abelian factor of order 2^k, the verifier checks k commuting involutions, literal generation, and the exact closure size 2^k. The induced surjection $\mathrm{C}_2^k \to H$ therefore is an isomorphism. This applies in particular to the new $k = 5, 6$ occurrences. Cyclic, dihedral and other previously supported atoms use their acquired complete normal-form and literal relation checks.

Candidate discovery is not a premise. The first attempt stopped at an unsupported nonsolvable complement calculation; the second reached its bounded deadline. Both are preserved without success credit. The successful successor additionally proposes a centralizer as a direct factor and caches failed small-group decomposition proposals. These operations supply candidates only. The independent reader still checks every successful factor, map and row by the preceding tests, so neither a cache miss nor an incorrect proposal can validate a false label.

Validation and scope. The ordinary and optimized Python checks verify all 496 row joins and every new model and map. Damaged source trees, actual embeddings, named top inputs, action degrees, and malformed source syntax are rejected. Independent elementary abelian examples of orders $32, 64$, actual split and nonsplit index-two examples, and intersecting purported direct factors exercise the elementary boundaries. The exact earlier registry proofs and reviews retain their attribution. This component supplies only these 496 structural descriptions; it makes no new historical census, timing, question, or whole-source claim.